Try 78 and Try 79 Formula Priors as Reused in Try 80: A Detailed Mathematical and Calibration Reference

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1 Purpose of this note

This note summarizes, in supervisor-facing form, the non-deep-learning priors that are reused in `Try 80`. The goal is to make clear:

- what comes directly from the literature,
- what was changed or adapted for the CKM dataset,
- what is inherited from earlier tries,
- **how every parameter was fitted in practice** (calibration procedure, solver, data split, and representative fitted values),
- and what is genuinely new in the final `Try 80` approach.

The note follows the implementation currently consumed by `TFGEightiethTry80/src/priors_try80.py`. Where calibration parameters originate from `Try 78` and `Try 79`, the relevant source artifacts are the JSON calibration files stored under those project directories.

2 High-level architecture

The central design decision in `Try 80` is *not* to learn path loss, delay spread, and angular spread from scratch. Instead:

- `Try 78` provides a frozen prior for `path_loss`,
- `Try 79` provides frozen priors for `delay_spread` and `angular_spread`,
- `Try 80` learns bounded residual corrections around those priors.

Conceptually, this means that the deterministic or interpretable structure is kept in the prior, and the deep model is used only for the residual part that the prior cannot explain. Figure 1 illustrates the overall pipeline.

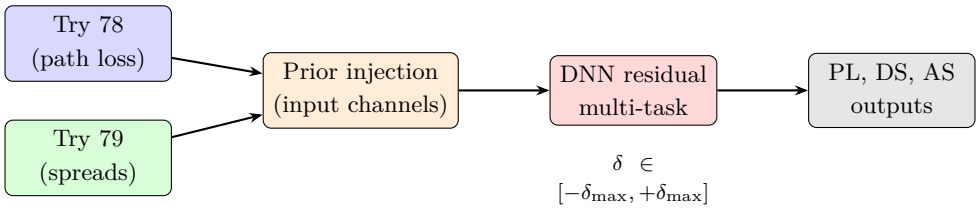

Figure 1: Overall Try 80 pipeline. The frozen priors from Try 78 and Try 79 are injected as input channels; the deep model predicts bounded residuals on top of them.

3 What was reused from the literature, what was changed, and what is new

This section gives a detailed, item-by-item account of (a) what the literature provides, (b) what we took from it, (c) what we deliberately changed or did *not* take, and (d) what is genuinely new in the combination.

3.1 Try 78: path loss

3.1.1 What the literature provides

Free-space path loss (FSPL). FSPL is a textbook formula [7] that gives the power loss of a plane wave in free space as a function of distance and frequency (Eq. 3). It is the universal starting point for any distance-dependent path-loss model. There is nothing new in using it.

Two-ray / ground-reflection model. The two-ray model for a transmitter above a flat reflecting ground is also classical [7]. For terrestrial base stations, the "two-ray" approximation is standard at distances beyond the breakpoint $d_{\mathrm{bp}} = 4h_{\mathrm{tx}}h_{\mathrm{rx}}/\lambda$, where the coherent sum of the two rays simplifies to a d^{-4} power law. For UAV air-to-ground links at moderate distances, Su et al. [3] and Khawaja et al. [4] both confirm that two-ray behavior is observable and produces the characteristic ring-like interference pattern.

COST-231 / Hata urban path loss. The COST-231 extension of the Hata model is a standard empirical formula for macrocell path loss in the 1.5–2 GHz range and was later applied at higher frequencies. It captures the dependence on base-station height, distance, and urban morphology class. It is used as a coarse NLoS baseline in many system-level studies.

LoS/NLoS regime separation. Treating LoS and NLoS as fundamentally different propagation regimes is a standard practice in 3GPP TR 38.901 [1], in UAV channel models [4], and in the A2G model family.

3.1.2 What we took exactly from the literature

- The FSPL formula (Eq. 3) is used verbatim, without any modification to the constants or exponents.
- The two-ray *concept* — direct ray plus ground-reflected ray, with a complex-valued sum — is taken directly from the standard derivation in [7, 3].
- The COST-231 formula (Eq. 18) is used verbatim, including the empirical constants 46.3, 33.9, 13.82, 44.9, 6.55, and the +3 dB metropolitan correction.
- The qualitative idea of LoS/NLoS regime separation is taken from [1, 4].

3.1.3 What we changed or did not take

- **Reflection coefficient Γ.** In the standard two-ray model, Γ is computed from Fresnel equations for a known ground material (permittivity, conductivity). We do *not* use material-based Fresnel coefficients. Instead, Γ is replaced by an *effective* complex parameter $\rho(h_{\mathrm{tx}})\, e^{j\phi(h_{\mathrm{tx}})}$ fitted per UAV height bin from data. The motivation is that the "ground" in an urban CKM map is not a homogeneous flat surface, and the effective reflection depends on the mix of buildings, streets, and vegetation visible at each UAV height.
- **Height-bin fitting instead of a single model.** Literature two-ray models are typically written as a single formula for a given scenario. Here, independent (ρ, ϕ, b) triples are fitted per 5 m height bin (Section 4.2), producing a height-dependent effective model. This is

necessary because the CKM dataset covers a wide range of UAV heights (roughly 10–460 m) and the interference geometry changes substantially across that range.

- **Radial residual correction.** Standard two-ray models do not include a separate range-dependent correction term $r(d_{2D})$. We add this because inspection of the CKM LoS residuals after the two-ray fit revealed a systematic smooth trend as a function of distance, likely caused by the finite urban clutter at ground level that the idealized flat-earth two-ray ignores.
- **COST-231 application to UAV.** The COST-231 model was developed for terrestrial base stations; applying it to a UAV that may be 10–400 m above the ground is an extrapolation beyond its original validation range. We use it only as a coarse floor for the NLoS raw prior (Eq. 22), combined via max with the A2G envelope. It is never used as the final NLoS prediction on its own.
- **Regime-wise linear calibration on top of COST-231.** The literature does not provide a way to calibrate a coarse formula against local building morphology at pixel level. We add the 15-feature design vector $X_{78}(x)$ and regime-wise OLS calibration (Eq. 46–48) to correct the raw prior using density, height, NLoS support, and elevation features derived from the CKM map itself. This step has no direct literature equivalent.

3.1.4 What is genuinely new in Try 78

The contribution of Try 78 is the *combination* and the dataset-driven calibration, not any individual formula:

1. fitting an effective two-ray model per height bin from CKM data rather than using Fresnel-based material coefficients,
2. adding a smooth radial residual on top of the two-ray fit,
3. building a regime-wise calibrated NLoS model that uses local CKM morphology features (density, height, NLoS support) to go beyond what COST-231 alone provides,
4. and producing a single hybrid LoS/NLoS path-loss map that is frozen and reused as a prior in Try 80.

None of steps 1–4 individually invents a new propagation formula. Each one adapts a known tool to the CKM pixel-map prediction setting in a way that is not found as a ready-made combination in any single reference.

3.2 Try 79: delay spread and angular spread

3.2.1 What the literature provides

Log-normal / log-domain treatment of spreads. 3GPP TR 38.901 [1] and WINNER II [2] model delay spread and angular spread as log-normal random variables, i.e., their logarithms are Gaussian. This is empirically well-established across many measurements and scenarios. The mean of the log-normal varies with distance, height, and scenario class, and separate parameters are given for LoS and NLoS.

LoS/NLoS and scenario-class conditioning. Both 3GPP and WINNER models provide separate tables of spread parameters for LoS vs. NLoS conditions and for different scenario types (urban macro, urban micro, indoor, suburban, etc.). The LoS/NLoS distinction is fundamental: LoS channels have smaller, more concentrated angular spreads and shorter delay spreads because the direct ray dominates.

Distance and elevation-angle dependence. Empirical studies on UAV channels [5, 6] report that delay spread and angular spread increase as the UAV elevation angle decreases (i.e., at longer horizontal distances), consistent with the UAV seeing more buildings at shallow angles.

Positive support and right-skewed distributions. Both delay spread and angular spread are strictly positive. The literature universally uses log-domain or log-normal representations because linear-space Gaussian models would assign probability to negative values.

3.2.2 What we took exactly from the literature

- The log-domain representation: we model $\log(1 + y)$ rather than y directly (Eq. 52), in line with 3GPP and WINNER practice.
- The LoS/NLoS separation: different baseline levels and different calibrated coefficient vectors per propagation state.
- The dependence on distance and elevation angle: included as explicit terms $\log(1 + d_{2D})$ and θ_{inv} in both the raw prior (Eq. 54) and the design matrix X_{79}.
- The qualitative finding from [5, 6] that elevation angle is a key predictor of spread: encoded via θ_{inv} and its interactions.

3.2.3 What we changed or did not take

- **No stochastic channel simulator.** 3GPP TR 38.901 and WINNER II provide spread parameters for generating random channel realizations per scenario drop. We do *not* generate random realizations. Instead, we fit a deterministic regression that predicts the *expected* spread value at each map pixel, conditioned on the local geometry.
- **No generic scenario tables.** 3GPP gives different tables for UMa, UMi, InH, RMa, etc. We do not use those tables directly. The baseline coefficients (b_{LoS}, b_{NLoS}, $a_1, \ldots, a_6$ in Eq. 54) are set from CKM-specific inspection, not from a 3GPP lookup table.
- **Map-derived morphology features.** Literature models do not include per-pixel features such as local building density $\rho_k(x)$, mean building height $h_k(x)$, NLoS-support fraction $n_k(x)$, blocker depth $b_{41}(x)$, transmitter clearance $c_{41}(x)$, or taller-building fraction $t_{41}(x)$. These 11 spatial features in X_{79} (Eq. 55) are computed directly from the CKM map topology tensor and have no equivalent in 3GPP or WINNER.
- **Regime-wise ridge regression per pixel.** 3GPP provides per-scenario stochastic parameters, not per-pixel regression coefficients. The regime-wise ridge regression (Eq. 68) and the associated fallback hierarchy are entirely our construction, designed to handle the sparse-regime problem that arises when the dataset is stratified across metric $\times$ topology $\times$ LoS/NLoS $\times$ antenna_bin.

- **UAV height as a feature.** The normalized height h_{norm} and its square h_{norm}^2 (Eq. 56) are included because the CKM dataset spans a wide range of UAV heights, and spread varies substantially with height. Standard 3GPP models do not include a continuous UAV height dimension in this way.

3.2.4 What is genuinely new in Try 79

1. Building a raw log-domain prior from simple geometry (distance, elevation angle, LoS/NLoS) and then refining it with per-pixel map morphology features in a single linear framework.
2. Using a 23-feature design matrix that combines the raw prior, multiscale spatial density/height/NLoS features, and the new obstruction/clearance features (b_{41}, c_{41}, t_{41}) derived from the CKM topology tensor.
3. Fitting regime-wise ridge regression coefficients from the CKM training split with incremental sufficient-statistics accumulation, allowing the model to scale to the full dataset without ever materializing the full design matrix in memory.
4. The fallback hierarchy that gracefully degrades to less specific regimes when training data is sparse.

Again, no individual step is a new physical formula. The contribution is the *pipeline* that maps from a CKM map tensor to a spread prediction map in a way that is transparent, interpretable, and trainable.

4 Try 78: Path Loss Prior

4.1 LoS branch: coherent two-ray model

4.1.1 Geometry and distances

Consider a UAV transmitter at height h_{tx} (metres above the ground-plane reference) and a receiver at height h_{rx}. For each map pixel x at 2D horizontal distance $d_{2D}(x)$ from the transmitter, two propagation distances are defined.

The *direct-path* (LoS) distance is:

$$d_{\text{LoS}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} - h_{\text{rx}})^2} \tag{1}$$

This is simply the Euclidean distance between the two antennas in 3D space. Because both $d_{2D}(x)$ and the height difference $(h_{\text{tx}} - h_{\text{rx}})$ are non-negative reals, $d_{\text{LoS}}(x) \geq d_{2D}(x)$, with equality only when both antennas are at the same height.

The *reflected-path* (image) distance is:

$$d_{\text{ref}}(x) = \sqrt{d_{2D}(x)^2 + (h_{\text{tx}} + h_{\text{rx}})^2} \tag{2}$$

This is the distance from the image source (the UAV mirrored below the ground plane at $-h_{\text{tx}}$) to the receiver. Since $(h_{\text{tx}} + h_{\text{rx}}) \geq |h_{\text{tx}} - h_{\text{rx}}|$ for any non-negative heights, we always have $d_{\text{ref}}(x) \geq d_{\text{LoS}}(x)$, reflecting the fact that the reflected path is always longer than the direct path.

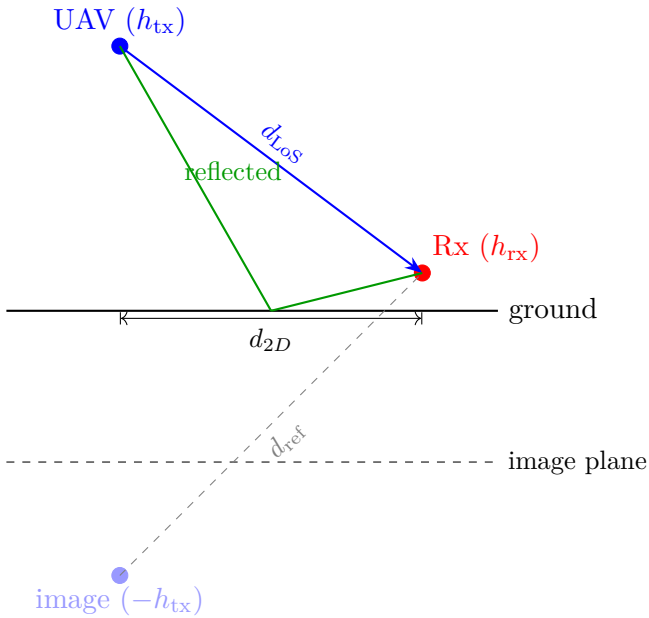

Figure 2: Two-ray geometry. Direct path (blue) has length d_{LoS}; reflected path (green) has the same total length as the image-source path (gray dashed), which is d_{ref}.

4.1.2 Free-space path loss

The free-space path loss in dB is defined using the standard formula at frequency f_{MHz} (in MHz):

$$\mathrm{FSPL}(x) = 32.45 + 20\log_{10}\left(\frac{d_{\mathrm{LoS}}(x)}{1000}\right) + 20\log_{10}(f_{\mathrm{MHz}}) \tag{3}$$

Derivation. FSPL in linear power ratio is $(4\pi d/\lambda)^2$ where $\lambda = c/f$. In dB:

$$\begin{aligned}
\mathrm{FSPL}_{\mathrm{dB}} &= 20\log_{10}\left(\frac{4\pi d f}{c}\right) \\
&= 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c) \\
&\approx 20\log_{10}(d_{\mathrm{km}} \cdot 1000) + 20\log_{10}(f_{\mathrm{MHz}} \cdot 10^6) + 20\log_{10}(4\pi) - 20\log_{10}(3 \times 10^8) \\
&= 20\log_{10}(d_{\mathrm{km}}) + 20\log_{10}(f_{\mathrm{MHz}}) + 32.45 \quad [\mathrm{dB}]
\end{aligned} \tag{4}$$

The constant $32.45 = 20\log_{10}(4\pi \cdot 10^3 \cdot 10^6/(3 \times 10^8)) \approx 32.45$ dB. In Eq. (3), $d_{\mathrm{LoS}}(x)/1000$ converts metres to kilometres.

Physical meaning. FSPL grows quadratically with distance ($\propto d^2$) and quadratically with frequency. It captures only the geometric spreading of the wavefront, with no ground interaction, scattering, or diffraction.

4.1.3 Coherent two-ray correction

In the two-ray model, the received field is the superposition of the direct path and a ground-reflected path. Let $k = 2\pi f/c$ be the free-space wavenumber. The complex sum of the two waves at the receiver is:

$$E_{\text{total}} \propto \frac{e^{-jkd_{\text{LoS}}}}{d_{\text{LoS}}} + \Gamma \frac{e^{-jkd_{\text{ref}}}}{d_{\text{ref}}} \tag{5}$$

where Γ is the complex ground reflection coefficient. In the standard textbook derivation, Γ is computed from the Fresnel equations for the ground material. Here, it is replaced by an *effective* complex parameter:

$$\Gamma(h_{\text{tx}}) = \rho(h_{\text{tx}})\, e^{j\phi(h_{\text{tx}})} \tag{6}$$

where $\rho \in [0, 1.5]$ is the effective amplitude (allowing slight over-unity to capture constructive multipath effects) and $\phi \in [-\pi, \pi]$ is the effective phase offset. Both depend on the UAV height bin.

The two-ray path-loss correction term in dB is:

$$C_{2\text{ray}}(x) = -20 \log_{10} \left| 1 + \rho(h_{\text{tx}}) \frac{d_{\text{LoS}}(x)}{d_{\text{ref}}(x)} e^{-j\left(k\left(d_{\text{ref}}(x) - d_{\text{LoS}}(x)\right) + \phi(h_{\text{tx}})\right)} \right| \tag{7}$$

Term-by-term interpretation.

- $\rho(h_{\text{tx}})$: amplitude of the ground reflection. A value of $\rho = 0$ recovers pure FSPL. The fitted values are $\rho \approx 0.49$–0.56 for heights 10–100 m, consistent with a moderate ground-reflection coefficient.
- $d_{\text{LoS}}(x)/d_{\text{ref}}(x)$: path amplitude ratio. This is always ≤ 1 since the reflected path is longer; it corrects for the $1/d$ amplitude decay of the reflected ray relative to the direct ray.
- $k(d_{\text{ref}}(x) - d_{\text{LoS}}(x))$: the phase difference accumulated due to the extra path length of the reflected wave.
- $\phi(h_{\text{tx}})$: an additional phase offset capturing phase shifts upon reflection (e.g., phase shift at the boundary). The fitted values are $\phi \approx 0$ for all height bins, meaning in-phase reflection dominates.
- $-20 \log_{10} |\cdot|$: the total field amplitude at the receiver, normalized to the direct-path field. When the two waves add in phase, the field amplitude is > 1, giving a *negative* correction (the signal is *stronger* than FSPL alone); when they cancel, the correction is *positive* (fading loss).

This spatial interference pattern creates the characteristic ring-like behavior in the LoS path-loss maps: concentric rings of constructive and destructive interference around the transmitter.

4.1.4 Calibrated LoS prior

The full calibrated LoS prior is:

$$\text{PL}_{\text{LoS}}(x) = \text{FSPL}(x) + C_{2\text{ray}}(x) + b(h_{\text{tx}}) + r(d_{2D}(x), h_{\text{tx}}) \tag{8}$$

where:

- $b(h_{\text{tx}})$ is a height-dependent bias (dB): a constant correction that shifts the overall two-ray prediction for a given UAV height.

- $r(d_{2D}(x), h_{\text{tx}})$ is a radial residual correction: a smooth function of horizontal distance (and height bin) that captures systematic over-/under-prediction of the two-ray model as a function of range. It is obtained by smoothing the empirical mean residual in each distance bin after the two-ray fit.

The four terms in Eq. (8) are applied sequentially: FSPL gives the geometric baseline; the two-ray correction adds the interference pattern; the bias corrects for a systematic global offset per height; and the radial term corrects for range-dependent systematic errors.

4.2 Calibration of the LoS two-ray model

4.2.1 Data preparation

All training samples are used. For each sample, valid LoS pixels are selected as those satisfying:

1. topology label = 0 (ground pixel),
2. $m_{\text{LoS}}(x) > 0$ (marked as LoS),
3. the measured path loss is finite and ≥ 20 dB.

If a sample has more than 2 500 valid LoS pixels, a random subsample of 2 500 pixels is drawn (with a fixed seed). Samples are then grouped by UAV height into height bins of 5 m width:

$$h_{\text{bin}} = 5 \cdot \left\lfloor \frac{h_{\text{tx}}}{5} \right\rfloor \tag{9}$$

Each bin is further subsampled to a maximum of 8 000 pixels total.

4.2.2 Grid search for (ρ, ϕ)

For each height bin, (ρ, ϕ) are found by grid search:

$$\phi_{\text{grid}} = \left\{ -\pi + \frac{2\pi k}{N_\phi} : k = 0, 1, \ldots, N_\phi - 1 \right\}, \quad N_\phi = 48 \tag{10}$$

$$\rho_{\text{grid}} = \{0.00,\ 0.05,\ 0.10,\ \ldots,\ 1.50\} \tag{11}$$

For each pair $(\hat{\rho}, \hat{\phi}) \in \rho_{\text{grid}} \times \phi_{\text{grid}}$, and for each pixel x_i in the height bin, define the two-ray prediction without bias:

$$\hat{p}_i(\hat{\rho}, \hat{\phi}) = \text{FSPL}(x_i) - 20 \log_{10} \left| 1 + \hat{\rho}\, \frac{d_{\text{LoS}}(x_i)}{d_{\text{ref}}(x_i)} e^{-j\left(k(d_{\text{ref}} - d_{\text{LoS}}) + \hat{\phi}\right)} \right| \tag{12}$$

The optimal bias for each $(\hat{\rho}, \hat{\phi})$ pair is solved analytically as the empirical mean of residuals:

$$\hat{b}(\hat{\rho}, \hat{\phi}) = \frac{1}{N} \sum_{i=1}^{N} \Big(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) \Big) \tag{13}$$

where y_i is the measured path loss at pixel x_i and N is the number of pixels in the bin. The biased prediction is $\hat{p}_i + \hat{b}$, and the RMSE for the pair is:

$$\mathrm{RMSE}(\hat{\rho}, \hat{\phi}) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \Big(y_i - \hat{p}_i(\hat{\rho}, \hat{\phi}) - \hat{b}(\hat{\rho}, \hat{\phi}) \Big)^2} \tag{14}$$

The best pair is:

$$(\rho^*, \phi^*) = \arg\min_{(\hat{\rho}, \hat{\phi})} \mathrm{RMSE}(\hat{\rho}, \hat{\phi}) \tag{15}$$

4.2.3 Post-fit Gaussian smoothing across height bins

After fitting all bins independently, a Gaussian-weighted smooth is applied to stabilize sparse high-altitude bins. Let $\{h_k\}$ be the fitted height bin centers, $\{(\rho_k^*, b_k^*)\}$ the per-bin fitted values, and $\{N_k\}$ the pixel counts used in fitting. The smoothed values are:

$$\tilde{\rho}(h) = \frac{\sum_k w_k(h)\, N_k\, \rho_k^*}{\sum_k w_k(h)\, N_k}, \qquad w_k(h) = \exp\left(-\frac{(h - h_k)^2}{2\sigma_h^2} \right), \quad \sigma_h = 10 \text{ m} \tag{16}$$

and similarly for b. The phase ϕ is always replaced with its smoothed version (treating it as effectively constant across heights).

Bins where both ρ_k^* and b_k^* differ noticeably from the smoothed global trend are flagged as suspicious (typically high-altitude bins with few samples), and their raw fit values are replaced by the smoothed values.

4.2.4 Radial residual correction

After fitting the two-ray model, a residual correction is computed:

$$r(d_{2D}) = \mathrm{GaussSmooth}_{\sigma_r} \left(\mathrm{clip} \left(\mu_\Delta(d_{2D}),\, -2.5,\, +2.5 \right) \right) \tag{17}$$

where $\mu_\Delta(d_{2D})$ is the empirical mean of $(y - \hat{y}_{2\mathrm{ray}})$ averaged over all pixels at horizontal distance d_{2D}, $\sigma_r = 1.5$ pixels is a Gaussian smoothing radius, and clipping limits the correction to ± 2.5 dB to prevent over-fitting on noisy radius bins.

4.2.5 Fitted parameter values

Table 1 shows representative fitted LoS two-ray parameters across height bins.

Table 1: Representative fitted LoS two-ray parameters by height bin.

h_{tx} (m)	ρ^*	ϕ^* (rad)	b^* (dB)
10	0.5605	0.000	−0.037
20	0.5242	0.000	−0.042
30	0.4950	0.000	−0.046
50	0.5468	0.000	−0.014
80	0.5482	0.000	−0.027
100	0.5463	0.000	−0.017
140	0.5414	0.000	+0.255
200	0.5435	0.000	+0.561
460	0.8966	−0.047	+1.521

The dominant pattern is $\rho \approx 0.49$–0.56 with $\phi \approx 0$ for heights 10–100 m, meaning a moderate-amplitude in-phase ground reflection. Biases are near-zero (< 0.05 dB) for these bins. At very high altitudes (> 140 m), the increasing positive bias and higher ρ values reflect sparser training data; those bins are stabilized by Gaussian smoothing.

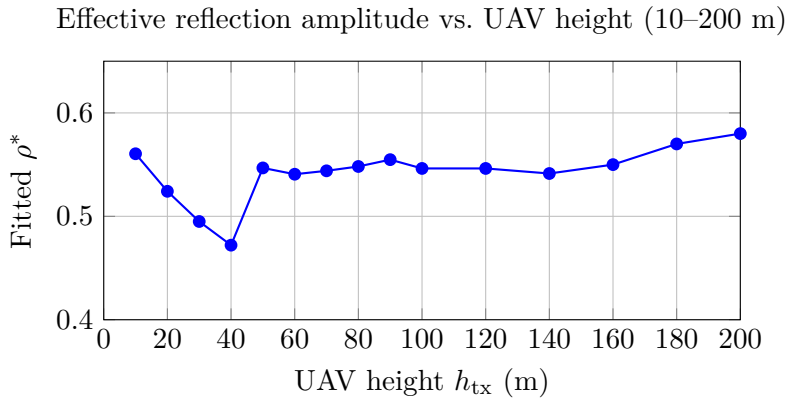

Figure 3: Effective ground-reflection amplitude ρ^* as a function of UAV height, fitted independently per 5 m height bin. Values are stable in the range 0.47–0.56 for 10–130 m, then rise slightly at higher altitudes where training data is sparser.

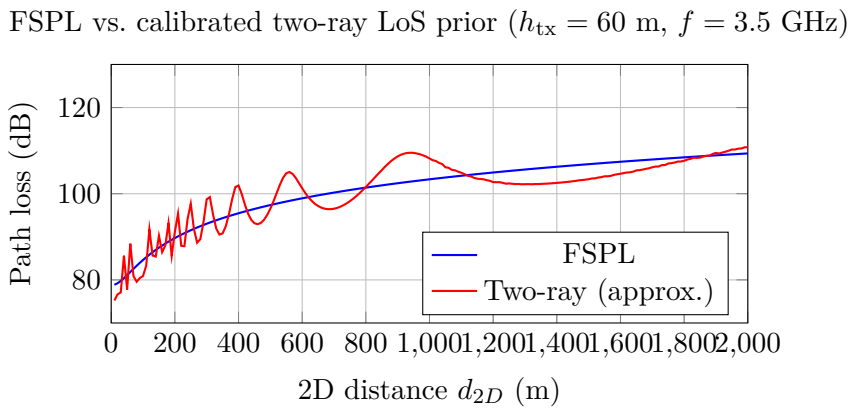

Figure 4: Schematic comparison of FSPL and the coherent two-ray model for a 60 m UAV at 3.5 GHz ($h_{\mathrm{rx}} = 1.5$ m). The two-ray model produces a characteristic oscillatory pattern around the FSPL baseline, corresponding to alternating zones of constructive and destructive interference.

4.3 NLoS branch

The NLoS path-loss branch used in `Try 80` is not a pure literature formula. It is a hybrid engineering model built in four stages.

4.3.1 Stage 1: raw NLoS propagation prior

Two coarse physical estimates are combined.

COST-231 term. The COST-231 / Hata extension for urban macro cells gives:

$$\mathrm{PL}_{\mathrm{COST231}}(x) = 46.3 + 33.9\log_{10}(f_{\mathrm{MHz}}) - 13.82\log_{10}(h_{\mathrm{tx}}) - a(h_{\mathrm{rx}}) + (44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))\log_{10}(d_{\mathrm{km}}) + 3 \tag{18}$$

where $d_{\mathrm{km}} = d_{2D}(x)/1000$ and the mobile antenna correction factor for large cities is:

$$a(h_{\mathrm{rx}}) = 3.2(\log_{10}(11.75\,h_{\mathrm{rx}}))^2 - 4.97 \tag{19}$$

Interpretation of Eq. (18)**.**

- $46.3 + 33.9\log_{10}(f_{\mathrm{MHz}})$: frequency-dependent offset. The exponent 33.9 (compare to 20 for FSPL) reflects stronger frequency dependence in urban scattering.
- $-13.82\log_{10}(h_{\mathrm{tx}})$: as the base station / UAV gets higher, path loss decreases (better clearance above rooftops). The -13.82 coefficient is a fitted constant from the Hata empirical dataset.
- $-a(h_{\mathrm{rx}})$: accounts for the receiver height. For typical pedestrian heights ($h_{\mathrm{rx}} \approx 1.5$ m), $a \approx -0.6$ dB, which is small.
- $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))\log_{10}(d_{\mathrm{km}})$: distance-dependent term. The effective path-loss exponent is $(44.9 - 6.55\log_{10}(h_{\mathrm{tx}}))/10$, which decreases as the UAV climbs (effectively reducing NLoS scattering losses at longer distances from higher altitudes).
- $+3$: the COST-231 metropolitan correction constant (dB).

Elevation-aware UAV envelope. The A2G NLoS envelope models the elevation-angle dependence of urban air-to-ground path loss:

$$\mathrm{PL}_{\mathrm{A2G,NLoS}}(x) = \lambda_0(h_{\mathrm{tx}}, f) + \beta_{\mathrm{NLoS}} + A_{\mathrm{NLoS}}\exp\left(-\frac{90 - \theta^\circ(x)}{\tau_{\mathrm{NLoS}}}\right) \tag{20}$$

where:

$$\lambda_0(h_{\mathrm{tx}}, f) = 20\log_{10}\left(\frac{4\pi h_{\mathrm{tx}} f}{c}\right) \tag{21}$$

The fitted constants are: $\beta_{\mathrm{NLoS}} = -16.16$ dB, $A_{\mathrm{NLoS}} = 12.0436$, $\tau_{\mathrm{NLoS}} = 7.52$ deg.

Interpretation of Eq. (20)**.**

- λ_0: a free-space-like offset that scales with UAV height and frequency [4, 3]. The CKM dataset uses $f = 7.125$ GHz (FR2 lower edge per 3GPP NR [1]); for $h_{\mathrm{tx}} = 50$ m at that frequency, $\lambda_0 \approx 103$ dB.
- $\beta_{\mathrm{NLoS}} = -16.16$ dB: a baseline NLoS attenuation offset relative to λ_0.
- $A_{\mathrm{NLoS}}\exp(-\ldots)$: an exponential modulation with $(90 - \theta^\circ)$. When the elevation angle θ° is high (UAV nearly overhead, $\theta^\circ \to 90^\circ$), the exponent vanishes and the term goes to A_{NLoS}. When θ° is low (grazing incidence), the exponent is large and the term approaches zero. The effect is that NLoS path loss *decreases* as the UAV moves overhead, consistent with improved clearance above buildings.
- $\tau_{\mathrm{NLoS}} = 7.52^\circ$: the characteristic angular scale. A low value means the transition from grazing to overhead happens sharply within a small elevation range.

The raw NLoS prior is then the conservative (higher) of the two estimates:

$$\mathrm{PL}_{\mathrm{NLoS,raw}}(x) = \max\left(\mathrm{PL}_{\mathrm{COST231}}(x), \mathrm{PL}_{\mathrm{A2G,NLoS}}(x)\right) \tag{22}$$

The full raw prior blends the LoS and NLoS estimates using the LoS mask $m_{\mathrm{LoS}}(x) \in [0, 1]$:

$$\mathrm{PL}^{(0)}(x) = m_{\mathrm{LoS}}(x)\,\mathrm{PL}_{\mathrm{LoS,blend}}(x) + (1 - m_{\mathrm{LoS}}(x))\,\mathrm{PL}_{\mathrm{NLoS,raw}}(x) \tag{23}$$

with a soft blend for the LoS branch:

$$\mathrm{PL}_{\mathrm{LoS,blend}}(x) = 0.7\,\mathrm{PL}_{\mathrm{LoS,path}}(x) + 0.3\min\left(\mathrm{PL}_{\mathrm{LoS,path}}(x), \mathrm{PL}_{\mathrm{A2G,LoS}}(x)\right) \tag{24}$$

The 0.7/0.3 blend in Eq. (24) gives 70% weight to the two-ray LoS model and 30% weight to the minimum of the two-ray model and an A2G LoS envelope. This prevents the prediction from exceeding a physically motivated upper bound in the LoS case.

4.3.2 Stage 2: map-derived morphology features

From $\mathrm{PL}^{(0)}(x)$ and the map topology, a 15-element feature vector is computed:

$$X_{78}(x) = \begin{bmatrix} \mathrm{PL}^{(0)}(x)^2 \\ \mathrm{PL}^{(0)}(x) \\ \log(1 + d_{2D}(x)) \\ \rho_{15}(x) \\ \rho_{41}(x) \\ h_{15}(x) \\ h_{41}(x) \\ \rho_{41}(x)\log(1 + d_{2D}(x)) \\ n_{15}(x) \\ n_{41}(x) \\ n_{41}(x)\log(1 + d_{2D}(x)) \\ \sigma_{\mathrm{shadow}}(x) \\ \theta_{\mathrm{norm}}(x) \\ n_{41}(x)\theta_{\mathrm{norm}}(x) \\ 1 \end{bmatrix} \tag{25}$$

Each term is now described in detail.

Base masks. The ground mask is:

$$g(x) = \begin{cases} 1, & \text{if topology}(x) = 0 \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

The building mask is $b(x) = 1 - g(x)$. The ground-NLoS support mask is:

$$n_{\mathrm{raw}}(x) = \mathbf{1}[m_{\mathrm{LoS}}(x) \leq 0.5] \cdot g(x) \tag{27}$$

This selects ground pixels that are predominantly in NLoS, i.e., pixels that are at street level and are shielded from the UAV.

Quadratic raw-prior feature. The pair $(\mathrm{PL}^{(0)}(x)^2, \mathrm{PL}^{(0)}(x))$ allows the linear model to fit a *quadratic* relationship in the raw prior. Including both P^2 and P as features lets the calibration curve be: $\hat{P} = \beta_1 P^2 + \beta_2 P + \cdots$, which can correct for systematic curvature in the raw prior.

Distance term. $\log(1 + d_{2D}(x))$: the logarithm of one plus the horizontal distance, providing a distance-dependent offset. The $+1$ prevents a singularity at $d_{2D} = 0$.

Per-pixel box-mean operator. All multiscale features use the same 2D box-mean operator $\mathrm{BoxMean}_k$. For a map $f : \mathbb{Z}^2 \to \mathbb{R}$ and a window half-width $p = \lfloor k/2 \rfloor$, the value at pixel $x = (i, j)$ is:

$$\mathrm{BoxMean}_k(f)(i,j) = \frac{1}{k^2} \sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\mathrm{padded}}(i', j') \tag{28}$$

where f_{padded} is f extended outside the map boundary by *reflect* (mirror) padding: $f_{\mathrm{padded}}(-r, j) = f_{\mathrm{padded}}(r, j)$ and similarly at all four edges. This ensures border pixels receive a full k^2-pixel average, with the map reflected back inward rather than zero-padded.

Equation (28) is evaluated efficiently using a *2D summed-area table* (integral image). Let $I(i, j) = \sum_{i' \le i, j' \le j} f_{\mathrm{padded}}(i', j')$ be the prefix-sum array. Then:

$$\sum_{i'=i-p}^{i+p} \sum_{j'=j-p}^{j+p} f_{\mathrm{padded}}(i', j') = I(i{+}p,\, j{+}p) - I(i{-}p{-}1,\, j{+}p) - I(i{+}p,\, j{-}p{-}1) + I(i{-}p{-}1,\, j{-}p{-}1) \tag{29}$$

This reduces each per-pixel evaluation to four table lookups, making the full-map computation $O(HW)$ regardless of k. In the implementation, $k \in \{15, 41\}$ pixels, corresponding to $\sim$15 m and $\sim$41 m neighborhoods (1 m/pixel resolution).

Multiscale building density.

$$\rho_k(x) = \mathrm{BoxMean}_k(b(\cdot))\,(x) = \frac{1}{k^2} \sum_{x' \in W_k(x)} b(x') \tag{30}$$

where $W_k(x)$ is the $k \times k$ window centered at x. Since $b(x') \in \{0, 1\}$, $\rho_k(x)$ is the *fraction of pixels that are buildings* in the window. It is zero at a ground pixel surrounded only by other ground pixels, and one at a pixel deep inside a solid building block. ρ_{15} captures fine-scale building density within a $\sim$15 m neighborhood; ρ_{41} captures a coarser $\sim$41 m neighborhood.

Multiscale building height.

$$h_k(x) = \frac{\mathrm{BoxMean}_k(\mathrm{topology}(\cdot)\, b(\cdot))\,(x)}{H_{\mathrm{norm}}}, \quad H_{\mathrm{norm}} = 90 \text{ m} \tag{31}$$

This is the box mean of $\mathrm{topology}(x') \cdot b(x')$: building pixels contribute their height in metres; ground pixels contribute 0. Dividing by k^2 (inside $\mathrm{BoxMean}_k$) gives:

$$h_k(x) = \frac{1}{H_{\mathrm{norm}}\, k^2} \sum_{x' \in W_k(x)} \mathrm{topology}(x') \cdot b(x') = \frac{\rho_k(x)}{H_{\mathrm{norm}}}\, \overline{h}_{\mathrm{bldg},k}(x) \tag{32}$$

where $\overline{h}_{\text{bldg},k}(x)$ is the *mean height of building pixels only* within the window:

$$\overline{h}_{\text{bldg},k}(x) = \frac{\sum_{x' \in W_k(x)} \text{topology}(x')\, b(x')}{\sum_{x' \in W_k(x)} b(x')} \quad \text{(undefined if } \rho_k(x) = 0) \tag{33}$$

Critically, $h_k(x) = \rho_k(x) \cdot \overline{h}_{\text{bldg},k}(x)/H_{\text{norm}}$, so h_k combines *both* how many buildings are present and how tall they are. A window with 50% building coverage and mean height 20 m gives the same h_k as a window with 100% coverage at mean height 10 m. The $H_{\text{norm}} = 90$ m normalizer maps the joint range to approximately $[0, 1]$.

Multiscale NLoS-support.

$$n_k(x) = \text{BoxMean}_k(n_{\text{raw}}(\cdot))\,(x) \tag{34}$$

Since $n_{\text{raw}}(x') \in \{0, 1\}$ (Eq. 27), $n_k(x)$ is the *absolute fraction of the window area* that consists of ground pixels in NLoS. Just as $h_k(x)$ conflates building coverage and building height, $n_k(x)$ conflates ground coverage and NLoS probability. We can write:

$$n_k(x) = (1 - \rho_k(x))\, \overline{n}_{\text{ground},k}(x) \tag{35}$$

where $\overline{n}_{\text{ground},k}(x)$ is the relative fraction of *available ground pixels* that are shielded:

$$\overline{n}_{\text{ground},k}(x) = \frac{\sum_{x' \in W_k(x)} n_{\text{raw}}(x')}{\sum_{x' \in W_k(x)} g(x')} \quad \text{(undefined if } \rho_k(x) = 1) \tag{36}$$

Because building pixels never contribute to n_k, this feature is zero in completely open LoS areas (where $\overline{n}_{\text{ground},k} = 0$) *and* deep inside solid building blocks (where $1 - \rho_k = 0$). It reaches its maximum in narrow urban street canyons, where ground coverage is still significant but entirely shielded from the UAV.

Elevation angle.

$$\theta(x) = \arctan\left(\frac{h_{\text{tx}} - h_{\text{rx}}}{\max(d_{2D}(x), 1)}\right) \tag{37}$$

The $\max(\cdot, 1)$ prevents division by zero directly below the UAV. Then:

$$\theta_{\text{norm}}(x) = \text{clip}\left(\frac{\theta^\circ(x)}{90},\ 0,\ 1\right) \tag{38}$$

where $\theta^\circ = \theta \cdot 180/\pi$ is in degrees. $\theta_{\text{norm}} = 0$ at grazing incidence, $= 1$ directly overhead.

Shadowing proxy.

$$\sigma_{\text{LoS}}(x) = \alpha_{\text{LoS}}\,(90 - \theta^\circ(x))^{\mu_{\text{LoS}}} \tag{39}$$

$$\sigma_{\text{NLoS}}(x) = \alpha_{\text{NLoS}}\,(90 - \theta^\circ(x))^{\mu_{\text{NLoS}}} \tag{40}$$

with fitted values $\alpha_{\text{NLoS}} = 2.3197$, $\mu_{\text{NLoS}} = 0.2361$. The blended proxy is:

$$\sigma_{\text{shadow}}(x) = m_{\text{LoS}}(x)\,\sigma_{\text{LoS}}(x) + (1 - m_{\text{LoS}}(x))\,\sigma_{\text{NLoS}}(x) \tag{41}$$

This acts as a spatially varying uncertainty proxy: higher values at low elevation angles, where propagation conditions are more difficult.

Interaction terms. The three pairwise products $\rho_{41}(x)\log(1{+}d_{2D}(x))$, $n_{41}(x)\log(1{+}d_{2D}(x))$, and $n_{41}(x)\theta_{\text{norm}}(x)$ allow the linear model to capture effects where building density or NLoS support *modifies* the slope of path loss with distance or elevation angle, rather than just shifting it.

4.3.3 Stage 3: regime-wise linear calibration

The regime r is identified by three keys:

- *Coarse topology:* one of `open_lowrise`, `mixed_midrise`, or `dense_highrise`,
- *Propagation state:* `LoS` or `NLoS`,
- *Antenna height bin:* `low_ant`, `mid_ant`, or `high_ant`.

This gives up to $3 \times 2 \times 3 = 18$ regimes.

How topology class and antenna bin are computed. These are *per-map* scalar labels, not per-pixel quantities. They are derived once from the full topology tensor and UAV height.

The *map-level* building density and mean building height are:

$$\bar{\rho}_{\text{map}} = \frac{1}{HW}\sum_{x} b(x) \tag{42}$$

$$\bar{h}_{\text{map}} = \frac{\sum_{x} \text{topology}(x)\, b(x)}{\sum_{x} b(x)} \tag{43}$$

$\bar{\rho}_{\text{map}}$ is simply the fraction of map pixels that are buildings. $\bar{h}_{\text{map}}$ is the mean building height *over building pixels only* (the denominator excludes ground pixels). This is different from the per-pixel feature $h_k(x)$ in Eq. (32), which divides the same numerator by the full window area k^2 instead of by the number of building pixels.

The six topology classes are assigned by thresholding both quantities:

$$\text{class}_6 = \begin{cases} \texttt{open_sparse_lowrise} & \bar{\rho} \le d_1,\ \bar{h} \le h_1 \\ \texttt{open_sparse_vertical} & \bar{\rho} \le d_1,\ \bar{h} > h_1 \\ \texttt{dense_block_midrise} & \bar{\rho} \ge d_2,\ \bar{h} \le h_2 \\ \texttt{dense_block_highrise} & \bar{\rho} \ge d_2,\ \bar{h} > h_2 \\ \texttt{mixed_compact_lowrise} & d_1 < \bar{\rho} < d_2,\ \bar{h} \le h_1 \\ \texttt{mixed_compact_midrise} & d_1 < \bar{\rho} < d_2,\ \bar{h} > h_1 \end{cases} \tag{44}$$

with thresholds $d_1 = 0.12$, $d_2 = 0.28$ (density fractions), $h_1 = 12$ m, $h_2 = 28$ m. The three-class label (`open`/`mixed`/`dense`) is the prefix of class_6 and is used in `Try 80` reporting.

The antenna-height bin is assigned by UAV height tertiles:

$$\text{ant_bin} = \begin{cases} \texttt{low_ant} & h_{\text{tx}} \leq 58.12 \text{ m} \\ \texttt{mid_ant} & 58.12 < h_{\text{tx}} \leq 103.85 \text{ m} \\ \texttt{high_ant} & h_{\text{tx}} > 103.85 \text{ m} \end{cases} \tag{45}$$

The calibrated prediction for regime r is:

$$\widehat{\text{PL}}_r(x) = X_{78}(x)^\top \beta_r^{78} = \sum_{j=1}^{15} x_j(x)\, \beta_{r,j}^{78} \tag{46}$$

How β_r^{78} is fitted. Across all training samples whose regime label equals r, a linear regression is run:

$$\hat{\beta}_r^{78} = \arg\min_{\beta} \sum_{x \in \mathcal{S}_r} \Big(y(x) - X_{78}(x)^\top \beta \Big)^2 \tag{47}$$

where $\mathcal{S}_r$ is the set of all valid pixels in training samples belonging to regime r. This ordinary least squares problem has the closed form solution:

$$\hat{\beta}_r^{78} = \Big(\mathbf{X}_r^\top \mathbf{X}_r \Big)^{-1} \mathbf{X}_r^\top \mathbf{y}_r \tag{48}$$

where $\mathbf{X}_r$ stacks the feature vectors and $\mathbf{y}_r$ stacks the measured path-loss values. In practice, if the matrix is near-singular (sparse regime), a fallback to a less specific regime key is used (see the fallback chain in the Try 79 section, which uses the same strategy).

An example fitted coefficient vector for the `mixed_midrise|LoS|high_ant` regime is:

$$\beta_{\texttt{mixed|LoS|high}}^{78} = \begin{bmatrix} -0.000554,\ 0.761,\ 3.061,\ 0.127,\ 0.793,\ -2.318,\ -0.481,\ -0.157 \\ 0.155,\ -0.218,\ 0.060,\ -47.15,\ -28.08,\ -0.508,\ 48.83 \end{bmatrix}^\top$$

The large bias weight ($\beta_{15} = 48.83$) and the large negative shadowing-proxy weight ($\beta_{12} = -47.15$) partially cancel, together acting as a calibrated effective offset for this regime.

4.3.4 Stage 4: region replacement and clipping

$$\text{PL}_{\text{NLoS}}(x) = \text{clip}\Big(X_{78}(x)^\top \beta_r^{78}, 20, 180 \Big) \tag{49}$$

Hard clipping to $[20, 180]$ dB prevents the linear model from producing physically unreasonable values. The final hybrid path-loss prior is:

$$\text{PL}_{\text{prior}}(x) = \begin{cases} \text{PL}_{\text{LoS}}(x), & m_{\text{LoS}}(x) > 0.5 \\ \text{PL}_{\text{NLoS}}(x), & \text{otherwise} \end{cases} \tag{50}$$

4.4 Evaluation metric: pixel-weighted RMSE

All numerical results are reported as *pixel-weighted* (pw) RMSE or MAE. Building pixels (topology$(x) > 0$) are always excluded; only valid ground pixels with a finite, physically plausible target value contribute. Let $\mathcal{V}_k$ be the set of all such valid (s, x) pairs across all evaluation samples for task k. Then:

$$\mathrm{RMSE}_k^{\mathrm{pw}} = \sqrt{\frac{\sum_{(s,x)\in\mathcal{V}_k} \left(\hat{y}_k^{(s)}(x) - y_k^{(s)}(x)\right)^2}{|\mathcal{V}_k|}} \tag{51}$$

This is a single square root over all pooled valid pixels, not an average of per-sample RMSEs. Units are dB (path loss), ns (delay spread), and degrees (angular spread), all in the native non-transformed domain.

4.5 Observed Try 78 results

Table 2 summarizes the path-loss RMSE values obtained from the final hybrid evaluation.

Table 2: Try 78 pixel-weighted RMSE (dB) on the evaluation split. The "Role" column indicates whether the model is used as a frozen prior in `Try 80`, serves only as an ablation baseline, or is an intermediate construction step.

Model	RMSE (dB)	Role in Try 80
FSPL (LoS only)	3.9540	ablation baseline (not used)
FSPL + radial residual	3.7086	ablation baseline (not used)
Coherent two-ray (LoS)	1.7516	building block → folded into Hybrid LoS
Hybrid LoS	1.7516	**frozen LoS prior** (used in Try 80)
Hybrid NLoS	3.3967	**frozen NLoS prior** (used in Try 80)
Hybrid overall	**1.9246**	combined frozen prior (Try 80 input)

Why do Coherent two-ray (LoS) and Hybrid LoS show the same RMSE? The two rows report the same value (1.7516 dB) on the *evaluation split*, but they are not identical models.

- **Coherent two-ray (LoS)** is the standalone two-ray model (Eq. 8) applied only to LoS pixels, with no soft clipping from an A2G envelope.
- **Hybrid LoS** applies the blend of Eq. 24: $0.7\,\mathrm{PL}_{\mathrm{LoS,path}} + 0.3\min(\mathrm{PL}_{\mathrm{LoS,path}}, \mathrm{PL}_{\mathrm{A2G,LoS}})$. For the vast majority of pixels the two-ray prediction is already below the A2G LoS envelope, so the min resolves to the two-ray value and the blend collapses to $0.7P + 0.3P = P$. The two formulas therefore coincide on almost all evaluation pixels, producing the same aggregate RMSE.

The practical difference only surfaces on pixels where the two-ray model predicts *unrealistically low* path loss (constructive-interference peaks that overshoot the measurement). On those pixels the A2G envelope acts as a physical upper-bound cap, slightly raising the prediction and preventing it from being overly optimistic. On the training set, where such edge cases appear more frequently, Hybrid LoS was marginally better than the standalone two-ray; the improvement rounds away at the reported precision for the evaluation split. The hybrid formulation was therefore kept as the frozen LoS prior in `Try 80` because it is more conservative and physically bounded, even though the difference is not visible at this precision.

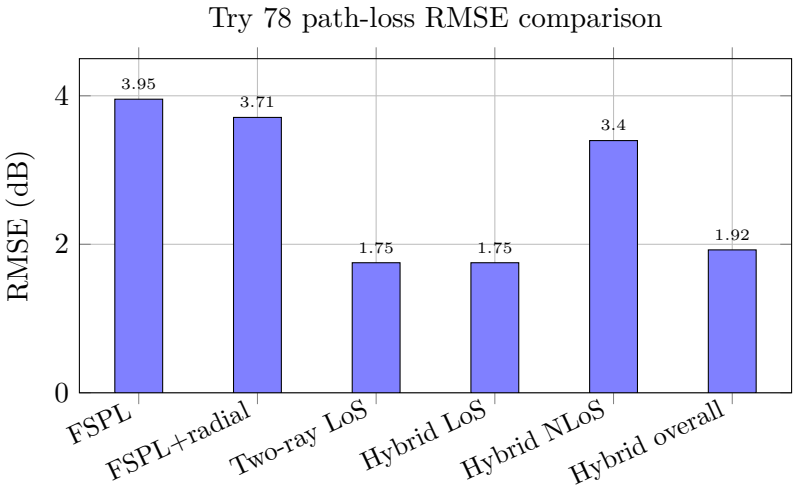

Figure 5: Try 78 pixel-weighted RMSE values. The coherent two-ray model reduces LoS RMSE from 3.95 dB (FSPL) to 1.75 dB, a gain of 2.2 dB.

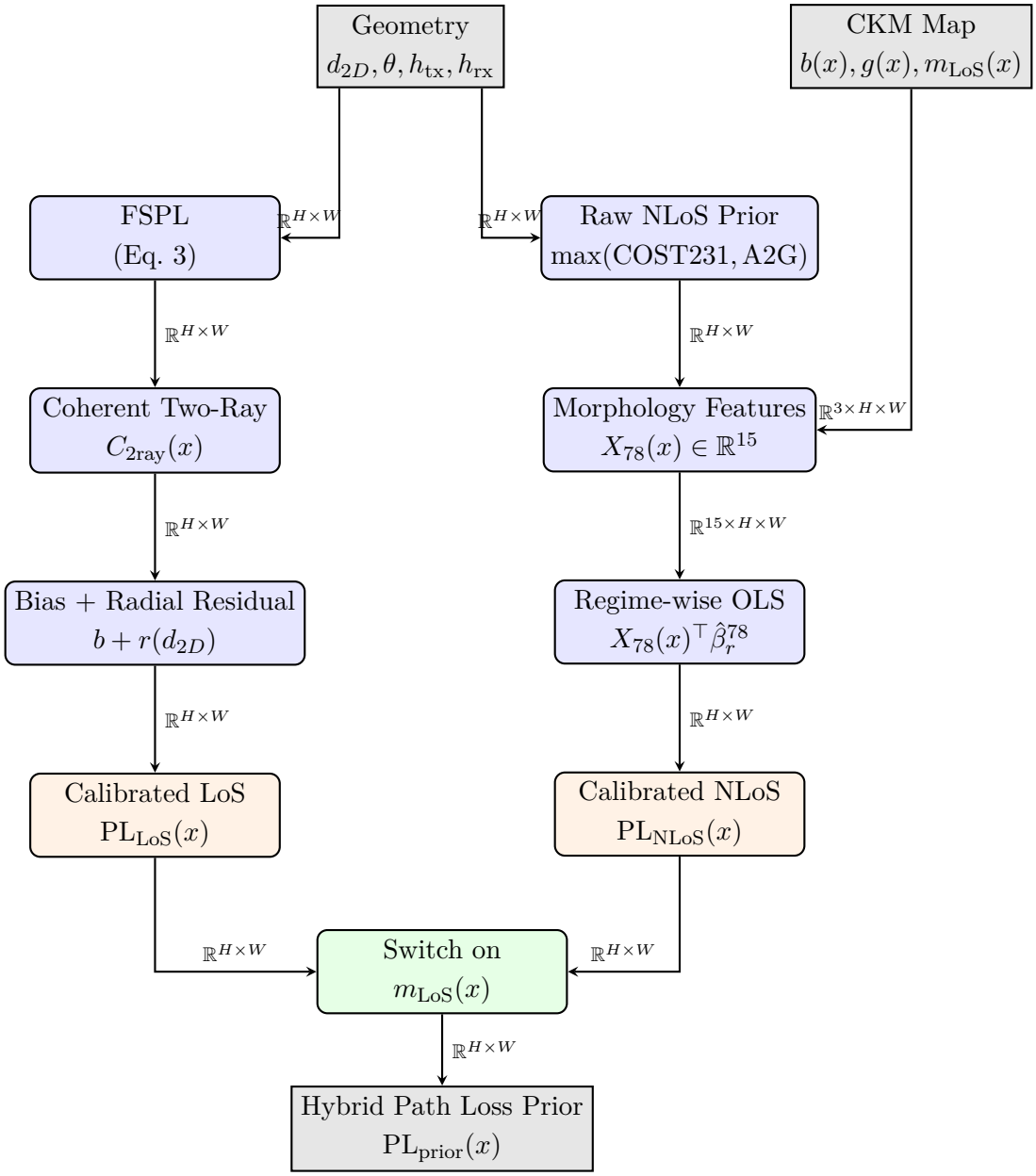

Figure 6: Pipeline for the `Try 78` hybrid path loss prior. The LoS branch relies on an explicit two-ray interference model, while the NLoS branch calibrates a raw prior against map-derived morphology features using regime-wise OLS. The two branches are hard-switched based on the LoS mask to produce the final frozen prior.

5 Try 79: Delay and Angular Spread Priors

5.1 Physical motivation for log-domain modeling

Delay spread (DS) and angular spread (AS) are strictly positive and heavy-tailed statistics. Their empirical distributions are typically log-normal, meaning $\log(1 + y)$ is approximately Gaussian [1, 2]. Modeling them in linear space would require non-negative regression, while the log transform allows standard linear regression tools to be applied directly.

For a spread variable $y \in \{\text{DS}, \text{AS}\}$, define:

$$z(x) = \log(1 + y(x)) \tag{52}$$

The +1 shift ensures $z \geq 0$ even when $y = 0$. The model is built to predict $z(x)$ and then the prediction is inverted via:

$$\hat{y}(x) = \exp(\hat{z}(x)) - 1 \tag{53}$$

5.2 Raw spread prior in log domain

The raw log-domain prior is built as a linear combination of geometric and morphological features:

$$z^{(0)}(x) = b_{\text{LoS/NLoS}}(x) + b_{\text{topology}} + a_1 \log(1+d_{2D}(x)) + a_2\theta_{\text{inv}}(x) + a_3\rho_{41}(x) + a_4 h_{41}(x) + a_5 n_{41}(x) + a_6 n_{41}(x)\theta_{\text{inv}}(x) \tag{54}$$

where $\theta_{\text{inv}}(x) = 1 - \theta_{\text{norm}}(x)$ (high value at low elevation, i.e., large off-zenith angle).

Interpretation of each term.

- $b_{\text{LoS/NLoS}}$: a baseline spread level that differs between LoS and NLoS. Typical values (in log-space): LoS base $\approx \log(1 + 4) = 1.61$ (4 ns for DS), NLoS base $\approx \log(1 + 11) = 2.49$ (11 ns). This captures the fact that NLoS channels are richer in multipath and therefore exhibit larger spreads.
- b_{topology}: a per-topology-class additive offset, capturing the fact that dense high-rise urban environments typically produce larger spreads than open sparse environments.
- $a_1 \log(1+d_{2D})$: distance-dependent spread growth. In 3GPP channel models, delay spread increases logarithmically with distance because farther scatterers contribute more delay [1].
- $a_2\theta_{\text{inv}}$: elevation-angle dependence. At low elevation angles (large θ_{inv}), the UAV sees more buildings at shallow angles, increasing both delay spread (more path length variation) and angular spread (wider scatterer angles).
- $a_3\rho_{41}$: higher building density increases spread.
- $a_4 h_{41}$: taller buildings produce longer path length differences, increasing delay spread.
- $a_5 n_{41}$: more NLoS ground pixels (more blocking) tends to increase spread (richer scattering environment).
- $a_6 n_{41}\theta_{\text{inv}}$: interaction: NLoS support has a stronger effect at low elevation angles.

Example raw prior coefficients for delay spread are: $b_{\text{LoS}} = 4.0$, $b_{\text{NLoS}} = 11.0$, $a_1 = 0.11$, $a_2 = 0.62$, $a_3 = 0.48$, $a_4 = 0.32$, $a_5 = 0.86$, $a_6 = 0.60$.

5.3 Calibrated spread prior

5.3.1 Stage 2: richer design matrix

The raw prior $z^{(0)}(x)$ is not the final estimate. A richer 23-element design vector is assembled:

$$X_{79}(x) = \begin{bmatrix} z^{(0)}(x)^2 \\ z^{(0)}(x) \\ \log(1 + d_{2D}(x)) \\ \theta_{\text{norm}}(x) \\ \theta_{\text{inv}}(x) \\ h_{\text{norm}} \\ h_{\text{norm}}^2 \\ \rho_{15}(x) \\ \rho_{41}(x) \\ h_{15}(x) \\ h_{41}(x) \\ n_{15}(x) \\ n_{41}(x) \\ n_{41}(x) \log(1 + d_{2D}(x)) \\ \rho_{41}(x)\theta_{\text{inv}}(x) \\ b_{41}(x) \\ 1 \\ c_{41}(x) \\ t_{41}(x) \\ \theta_{\text{norm}}(x)\rho_{41}(x) \\ c_{41}(x)\theta_{\text{inv}}(x) \\ t_{41}(x)\rho_{41}(x) \\ t_{41}(x)n_{41}(x) \end{bmatrix} \tag{55}$$

The new features (beyond those already in X_{78}) are:

Normalized transmitter height.

$$h_{\text{norm}} = \frac{\log(1 + \max(h_{\text{tx}} - h_{\text{rx}}, 0))}{\log(401)} \tag{56}$$

This compresses the UAV height into $[0, 1]$ on a log scale; the denominator $\log(401)$ normalizes by the maximum expected height difference (400 m). Both h_{norm} and h_{norm}^2 are included to allow a quadratic relationship between UAV height and spread.

Blocker depth.

$$b_{41}(x) = \frac{\text{BoxMean}_{41}(\max(\text{topology}(\cdot) - h_{\text{rx}}, 0)\, b(\cdot))\,(x)}{H_{\text{norm}}} \tag{57}$$

This measures how much buildings in the 41×41 window *protrude* above the receiver height. A large b_{41} means there are many tall buildings nearby, which act as diffraction edges and scatterers.

Transmitter clearance.

$$\text{clearance}(x) = \max(h_{\text{tx}} - \text{topology}(x), 0)\, b(x) \tag{58}$$

$$c_{41}(x) = \frac{\mathrm{BoxMean}_{41}(\mathrm{clearance}(\cdot))\,(x)}{H_{\mathrm{norm}}} \tag{59}$$

This measures how far the UAV is *above* surrounding buildings, i.e., the mean altitude clearance of the UAV over nearby building tops. A large c_{41} means the UAV is well above local rooftops, leading to better LoS and typically smaller spread.

Taller-building fraction.

$$t_{\mathrm{raw}}(x) = \mathbf{1}[\mathrm{topology}(x) > h_{\mathrm{tx}}] \cdot b(x) \tag{60}$$

$$t_{41}(x) = \mathrm{BoxMean}_{41}(t_{\mathrm{raw}}(\cdot))\,(x) \tag{61}$$

t_{41} is the fraction of buildings in the 41×41 window that are taller than the UAV. A high t_{41} means the UAV is operating *below* rooftop level and is more severely obstructed.

Interaction terms. Six pairwise interaction terms are included:

$$n_{41}(x)\log(1 + d_{2D}(x)) \quad \text{(NLoS support vs. distance)} \tag{62}$$

$$\rho_{41}(x)\theta_{\mathrm{inv}}(x) \quad \text{(density vs. elevation)} \tag{63}$$

$$\theta_{\mathrm{norm}}(x)\rho_{41}(x) \quad \text{(elevation vs. density, complement)} \tag{64}$$

$$c_{41}(x)\theta_{\mathrm{inv}}(x) \quad \text{(clearance vs. elevation)} \tag{65}$$

$$t_{41}(x)\rho_{41}(x) \quad \text{(taller-buildings vs. density)} \tag{66}$$

$$t_{41}(x)n_{41}(x) \quad \text{(taller-buildings vs. NLoS support)} \tag{67}$$

These allow the linear model to express, e.g., that the effect of building density on spread depends on whether the UAV is at low or high elevation.

5.3.2 Stage 3: regime-wise ridge regression

The regime r for Try 79 is keyed by:

- metric (`delay_spread` or `angular_spread`),
- topology class (6 classes),
- propagation state (LoS or NLoS),
- antenna-height bin (low / mid / high).

This gives up to $2 \times 6 \times 2 \times 3 = 72$ regimes.

Calibration fitting algorithm. Across all training samples in regime r, valid pixels are selected and subsampled at rate $p = 0.02$ (2%). The feature matrix $\mathbf{X}_r$ and target vector $\mathbf{z}_r = \log(1 + \mathbf{y}_r)$ (clipped to $[0, 8]$) are accumulated. If the regime has at least $4 \times 23 = 92$ pixels, the ridge regression is:

$$\hat{\beta}_r^{79} = \left(\mathbf{X}_r^\top \mathbf{X}_r + \lambda I_{23}\right)^{-1} \mathbf{X}_r^\top \mathbf{z}_r, \quad \lambda = 10^{-3} \tag{68}$$

The small ridge penalty $\lambda = 10^{-3}$ prevents numerical instability when the design matrix is near-singular (e.g., when a 41×41 box feature is nearly constant across a sparse regime).

Normal equations. The normal equations for ridge regression are:

$$\left(\mathbf{X}_r^\top \mathbf{X}_r + \lambda I\right) \hat{\beta}_r^{79} = \mathbf{X}_r^\top \mathbf{z}_r \tag{69}$$

This is solved by Cholesky decomposition, with a fallback to `numpy.linalg.lstsq` (minimum-norm least squares) if the matrix is still numerically singular.

Accumulation trick for large datasets. Since the CKM dataset contains ∼10 840 training samples and each sample contributes $O(10^4)$ pixels, forming the full $\mathbf{X}_r$ matrix in memory would be prohibitive. Instead, the sufficient statistics $\mathbf{X}_r^\top \mathbf{X}_r$ (23×23) and $\mathbf{X}_r^\top \mathbf{z}_r$ (23×1) are accumulated incrementally:

$$\mathbf{A}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{X}_{r,s} \quad \forall s \tag{70}$$

$$\mathbf{v}_r \mathrel{+}= \mathbf{X}_{r,s}^\top \mathbf{z}_{r,s} \quad \forall s \tag{71}$$

where s indexes training samples, and only $\mathbf{A}_r$ and $\mathbf{v}_r$ need to be stored. The ridge solution is then $\hat{\beta}_r^{79} = (\mathbf{A}_r + \lambda I)^{-1} \mathbf{v}_r$.

Fallback hierarchy. If a regime is too sparse to fit (< 92 pixels), the code falls back to progressively more general regimes:

1. exact regime: (metric, topology class, LoS/NLoS, antenna bin),
2. same topology class + LoS/NLoS + *all* antenna heights,
3. same topology class + *all* LoS states + all antenna heights,
4. global topology + LoS/NLoS + all antenna heights,
5. fully global fallback: (metric, global, all-LoS, all-ant).

This ensures that even data-sparse regimes always have a valid coefficient vector, at the cost of lower specificity.

Example fitted coefficients. For the regime `delay_spread|open_sparse_vertical|LoS|mid_ant`:

$$\hat{\beta}^{79} = \begin{bmatrix} -1.020, & 9.974, & -1.030, & -4.530, & -6.412, & 11.724, & -9.017, & -0.973, & -1.849, & -2.806, & -1.578, & -1.110 \\ & -6.950, & 0.540, & 0.521, & 6.621, & -10.929, & 0.267, & 3.567, & -2.367, & 1.058, & -15.276, & -3.614 \end{bmatrix}$$

The large positive weight on $z^{(0)}$ ($\beta_2 = 9.97$) and negative bias ($\beta_{17} = -10.93$) together calibrate the scale of the raw prior.

5.3.3 Stage 4: inverse transform and clipping

Once the calibrated log-domain prediction $\hat{z}(x) = X_{79}(x)^\top \hat{\beta}_r^{79}$ is obtained, it is mapped back to native units:

$$\hat{y}(x) = \exp\left(\hat{z}(x)\right) - 1 \tag{72}$$

Then regime-dependent clipping is applied:

$$\hat{y}(x) = \begin{cases} \text{clip}\left(\hat{y}(x), 0, y_{\max}^{\text{LoS}}\right), & m_{\text{LoS}}(x) = 1 \\ \text{clip}\left(\hat{y}(x), 0, y_{\max}^{\text{NLoS}}\right), & m_{\text{LoS}}(x) = 0 \end{cases} \tag{73}$$

The clip limits are:

- delay spread: $y_{\max}^{\text{LoS}} = y_{\max}^{\text{NLoS}} = 400$ ns,
- angular spread: $y_{\max}^{\text{LoS}} = 15°$, $y_{\max}^{\text{NLoS}} = 90°$.

The LoS angular spread clip of 15° reflects that in LoS conditions the direct ray dominates and the angular energy is concentrated near the direction of the UAV; in NLoS, the full 90° hemisphere is accessible via diffraction and scattering.

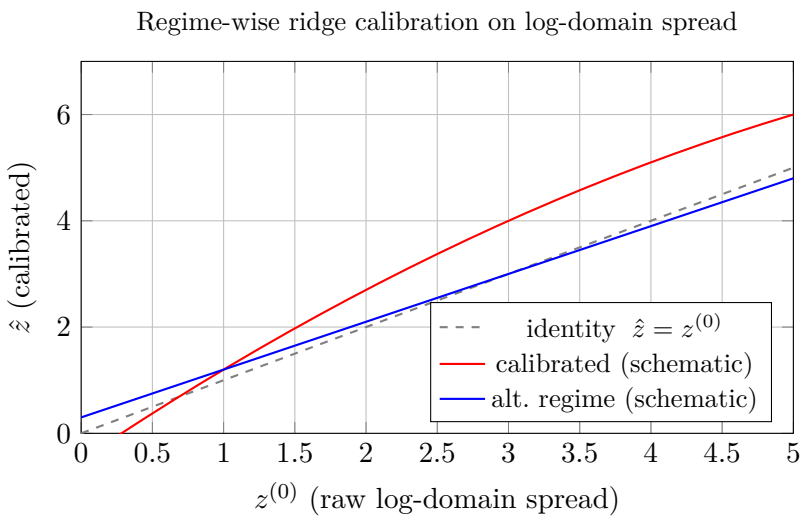

Figure 7: Schematic illustration of the regime-wise calibration. The raw log-domain prior $z^{(0)}$ (identity line) is mapped by the fitted linear model to the calibrated prediction $\hat{z}$. Different regimes produce different calibration curves, allowing the model to correct for systematic biases in the raw prior on a per-scene basis.

5.4 Observed Try 79 results

All values follow the pixel-weighted RMSE definition of Section 4.4: metrics are pooled across all valid *ground* pixels, buildings excluded, and a single square root is taken over the accumulated SSE.

5.4.1 Delay spread

Table 3: Delay spread: calibrated test RMSE (ns) by topology class.

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	16.8807	17.0146	15.4284
open_sparse_vertical	40.8366	40.4400	44.0772
mixed_compact_lowrise	17.2661	16.6524	18.9097
mixed_compact_midrise	37.8812	38.2608	37.2474
dense_block_midrise	26.9047	28.9951	25.0188
dense_block_highrise	43.5159	40.1619	45.0775
Overall	**28.1023**	**27.4851**	**29.6157**

Table 4: Delay spread: calibrated test RMSE (ns) by antenna-height bin.

Antenna bin	Overall	LoS	NLoS
low_ant	37.3435	39.7087	33.5247
mid_ant	29.0487	28.4214	30.5248
high_ant	12.2353	12.1513	12.6785

5.4.2 Angular spread

Table 5: Angular spread: calibrated test RMSE (°) by topology class.

Topology class	Overall	LoS	NLoS
open_sparse_lowrise	10.0141	10.3006	6.3311
open_sparse_vertical	12.6999	12.9825	9.9658
mixed_compact_lowrise	14.3745	15.9870	8.1715
mixed_compact_midrise	15.2580	17.8463	9.5741
dense_block_midrise	15.5326	21.2010	8.1714
dense_block_highrise	14.6372	21.3260	9.7809
Overall	**13.7416**	**15.2818**	**8.6614**

Table 6: Angular spread: calibrated test RMSE (°) by antenna-height bin.

Antenna bin	Overall	LoS	NLoS
low_ant	15.5124	18.4271	9.6818
mid_ant	14.4554	16.2652	8.5643
high_ant	10.8852	11.6030	5.6148

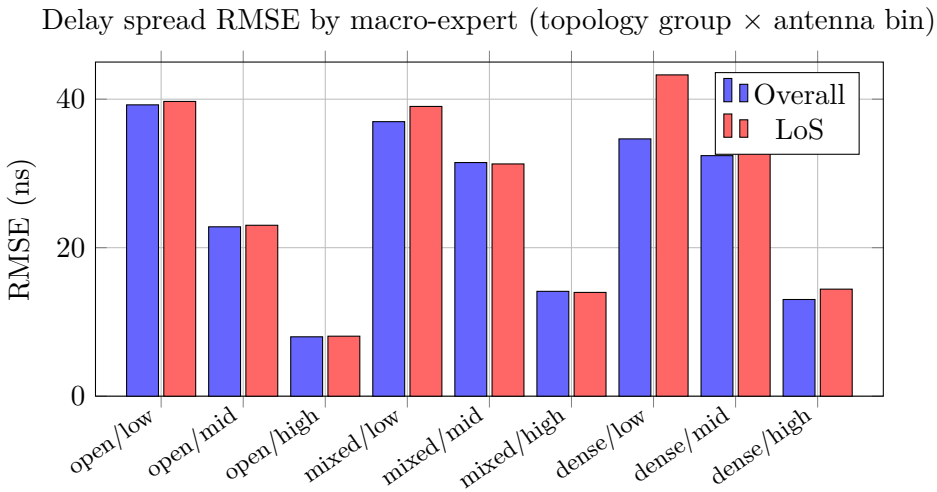

Figure 8: Delay spread RMSE (ns) broken down by topology macro-group and antenna-height bin. High-altitude UAVs (high_ant) consistently achieve the lowest RMSE across all topology groups.

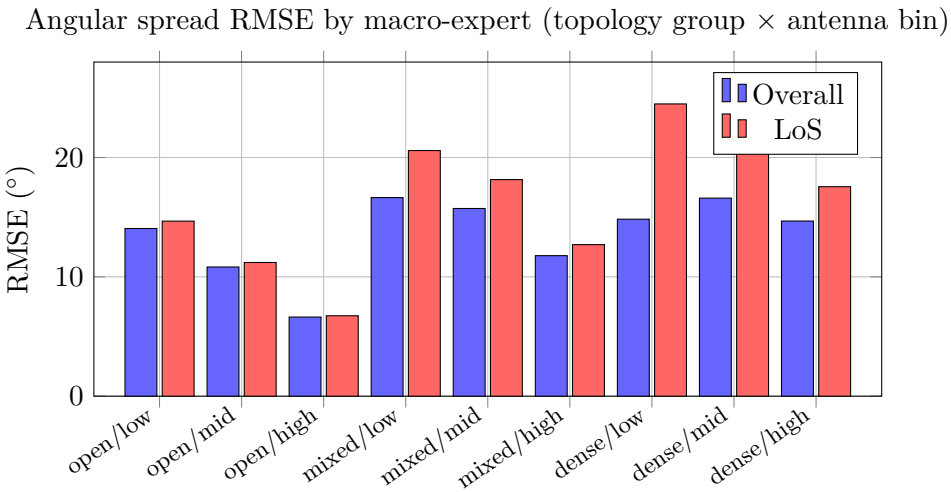

Figure 9: Angular spread RMSE (°) broken down by topology macro-group and antenna-height bin. The LoS RMSE is notably higher than NLoS RMSE in dense scenarios because LoS angular spread is dominated by a direct ray spike, which is harder to localize precisely.

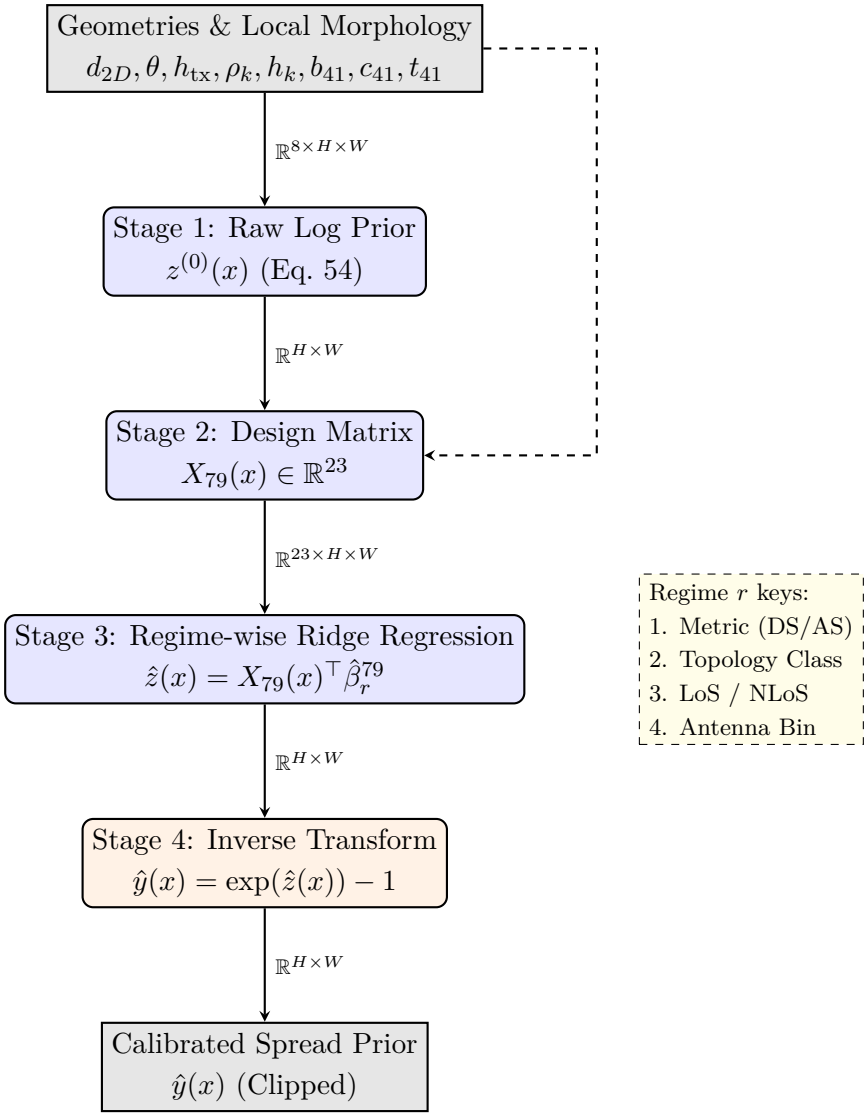

Figure 10: Pipeline for the `Try 79` delay and angular spread priors. Both spread modalities share this identical linear architecture. They predict the expected spread in the log domain using regime-wise ridge regression over 23 spatial and geometric features.

6 What is reused in Try 80, and what is new

6.1 What is not new in Try 80

The following are inherited and frozen:

- the calibrated two-ray LoS path-loss prior from `Try 78` (Eq. 8),
- the calibrated regime-wise NLoS path-loss prior from `Try 78` (Eq. 49),
- the calibrated delay-spread prior from `Try 79` (Eq. 72),
- the calibrated angular-spread prior from `Try 79` (Eq. 72).

6.2 What is new in Try 80

`Try 80` introduces a joint multi-task architecture that predicts a *Gaussian Mixture Model (GMM)* at each pixel, rather than a simple point estimate. For each task $k \in \{\text{PL, DS, AS}\}$,

the target value $y_k(x)$ is modeled as being drawn from a mixture of $J = 3$ components.

The mean of each mixture component j is anchored to the frozen prior:

$$\mu_{k,j}(x) = y_k^{\text{prior}}(x) + \alpha_k(x) \cdot \text{clip}\left(\delta_{\text{global},k,j} + \delta_{\text{local},k}(x), \pm\delta_{\text{max},k}\right) \tag{74}$$

where:

- $y_k^{\text{prior}}(x)$ is the frozen prior from `Try 78` or `Try 79`,
- $\delta_{\text{global},k,j}$ is a map-level residual offset for component j,
- $\delta_{\text{local},k}(x)$ is a per-pixel residual refinement shared across components,
- $\alpha_k(x) \in (0, 1)$ is a learned spatial gating factor,
- The sum is bounded to $\pm\delta_{\text{max},k}$ to prevent the DNN from deviating catastrophically from the prior.

The DNN also predicts the per-pixel mixture probabilities $p_{k,j}(x)$ and component variances $\sigma^2_{k,j}(x) = \sigma^2_{\text{global},k,j} + \tilde{\sigma}^2_{\text{local},k}(x)$.

The model is trained end-to-end to maximize the Negative Log Likelihood (NLL) of the measurements under this GMM. The deterministic point prediction used for RMSE evaluation is the expected value of the mixture:

$$\hat{y}_k(x) = \sum_{j=1}^{3} p_{k,j}(x)\mu_{k,j}(x) \tag{75}$$

The training loss also includes a prior-preservation term:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{prior}} \sum_k \mathbb{E}_x \left[\max\left(0,\ (\hat{y}_k(x) - y_k(x))^2 - \left(y_k^{\text{prior}}(x) - y_k(x)\right)^2\right)\right] \tag{76}$$

The second term is a *pixel-wise* prior-guard. It penalizes the model explicitly on any pixel where its squared error exceeds the prior's squared error. This asymmetric penalty is strictly stronger than an aggregate RMSE check, as it prevents the DNN from sacrificing performance in difficult local regions to optimize the global mean.

The four components of the Try 80 pipeline are therefore:

1. inject frozen priors as explicit map-shaped input channels,
2. multi-task DNN predicts a per-pixel GMM anchored to the prior,
3. train via NLL alongside a pixel-wise prior-preserving loss term,
4. final point prediction is the expected value of the mixture.

6.2.1 Architecture: Shared UNet with FiLM

Under the hood, `Try 80` processes the map inputs (such as building layouts and LoS/NLoS masks) through a shared multi-task convolutional encoder-decoder. Because the UAV altitude strongly modifies propagation paths, it is insufficient to simply append the scalar UAV height to the spatial input. Instead, the scalar UAV height is encoded into a condition vector $\mathbf{c} \in \mathbb{R}^{128}$ and injected dynamically into the convolutional feature maps at multiple scales via *Feature-wise Linear Modulation (FiLM)* layers.

For a given convolutional feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, a linear projection of the height condition $\mathbf{c}$ outputs a scale vector $\boldsymbol{\gamma} \in \mathbb{R}^C$ and a shift vector $\boldsymbol{\beta} \in \mathbb{R}^C$. The feature map is then modulated via an affine transformation:

$$\mathbf{X}' = \mathbf{X} \odot (1 + \boldsymbol{\gamma}) + \boldsymbol{\beta} \tag{77}$$

where $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are broadcasted spatially across the height and width. This mechanism allows the scalar altitude to globally scale and shift the activation of specific spatial filters—for instance, amplifying "far-field scattering" filters at high altitudes while suppressing them at low altitudes.

The architecture explicitly enforces the separation of global and local effects via a dual-head design for each task, directly mirroring the mathematical separation of (x)-dependent and non-(x)-dependent variables in Eq. 74:

- **Global Head (Map-Level Context)**: The deepest bottleneck feature map is globally average-pooled over its spatial dimensions to produce a single vector representing the macro-context of the entire map (e.g., "this is predominantly a dense high-rise environment"). This pooled vector, fused with the height condition, is passed through a Multi-Layer Perceptron (MLP) to predict the non-spatial quantities: mixture probabilities $p_{k,j}$, global offsets $\delta_{\text{global},k,j}$, and global variances $\sigma^2_{\text{global},k,j}$. These apply equally to all pixels, capturing the base scenario behavior.
- **Local Head (Pixel-Level Refinement)**: The high-resolution output of the decoder ($\mathbf{F} \in \mathbb{R}^{C' \times H \times W}$) is passed through 1×1 convolutions to predict the spatially varying fields. It outputs 1-channel maps for the residual refinement $\delta_{\text{local},k}(x)$, the spatial gating factor $\alpha_k(x)$, and the local variance $\tilde{\sigma}^2_{\text{local},k}(x)$. Because these terms are derived from the high-resolution spatial decoder branch, they react to micro-morphological corrections, like the specific placement of a building blocking the path.

This guarantees that broad contextual shifts (e.g., an entire map having systematically higher delay spread) are modeled separately from fine-grained morphological effects (e.g., a specific street canyon creating a localized multipath trap).

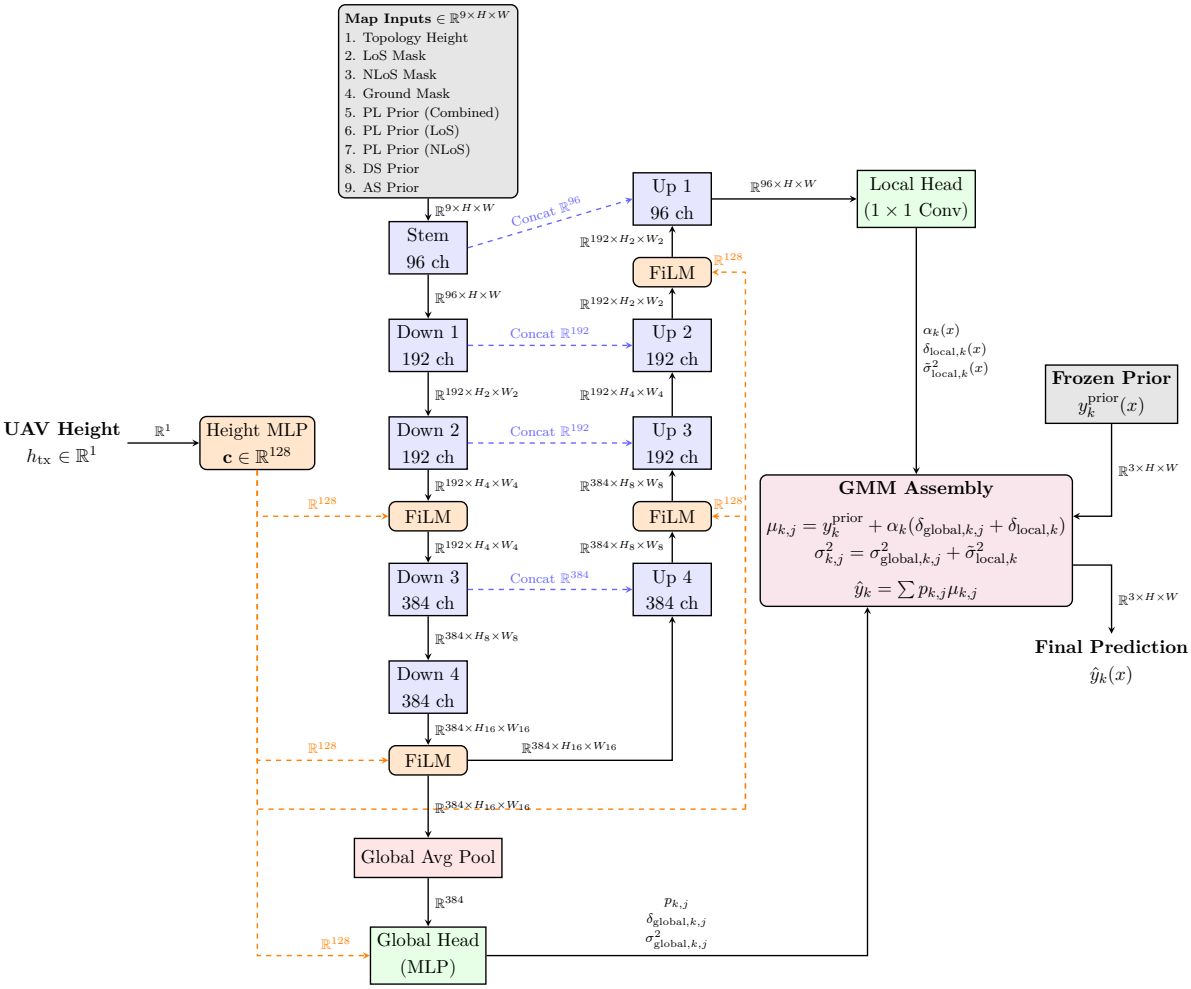

Figure 11: Schematic of the `Try 80` Shared UNet architecture with FiLM conditioning. The UAV height is encoded into a condition vector $\mathbf{c}$ and injected dynamically into multiple layers of the encoder and decoder. The architecture explicitly bifurcates into a Global Head (predicting map-level GMM constants from pooled features) and a Local Head (predicting per-pixel refinements from the high-resolution decoder output).

6.2.2 Composite Optimization Objective

Equation 76 highlights the core philosophy, but the actual implementation utilizes a robust composite loss to stabilize the GMM training and prevent mode collapse:

- **Negative Log-Likelihood** ($\mathcal{L}_{\text{NLL}}$): The primary data-fidelity term, computed under the predicted GMM and evaluated independently for LoS and NLoS regions using explicit region masks.
- **Prior Guard** ($\mathcal{L}_{\text{guard}}$): The asymmetric penalty (Eq. 76) that penalizes the network strictly on pixels where its squared error exceeds that of the deterministic physical prior.
- **Anchor Penalty** ($\mathcal{L}_{\text{anchor}}$): Direct L2 regularization on $\alpha_k(x)$, δ_{global}, and δ_{local}. This mathematically anchors the network to the physical priors, forcing the deep learning layers to "justify" any deviations.
- **Distributional KL-Divergence** ($\mathcal{L}_{\text{KL}}$): A soft-histogram loss that encourages the marginal distribution of the predicted GMM to match the empirical distribution of residuals in the current batch.

- **Outlier Budget**: A penalty that bounds the probability mass assigned to the third GMM component (designed for severe multipath outliers), preventing it from indiscriminately absorbing general variance.
- **Moment Matching & Auxiliary Losses**: Includes RMSE, MAE, and spatial moment-matching (penalizing differences in map mean and variance) applied directly to the deterministic expected value $\hat{y}_k(x)$ to accelerate early-stage convergence.

7 Why these changes were made

1. **The CKM maps show strong dataset-specific structure.** In particular, LoS path loss showed ring-like residual behavior around the transmitter, which justified a fitted two-ray treatment instead of a generic uncalibrated model.

2. **The task is pixel-map prediction, not only scenario-level modeling.** Classical channel models are usually written for statistical scenario simulation, whereas here the aim is to predict a full spatial field over a map.

3. **Interpretability was intentionally preserved.** Both `Try 78` and `Try 79` remain readable and coefficient-level inspectable. This makes them useful as thesis material and as robust baselines.

4. **Deep learning was introduced only after the priors were strong.** The philosophy in `Try 80` is that deep learning should model the residual uncertainty, not replace known deterministic structure.

8 Summary

The literature provided the physical and statistical starting points (two-ray LoS behavior, coarse urban path-loss envelopes, log-domain treatment of delay/angular spreads, and LoS/NLoS regime separation). The actual work was to adapt those ideas to the CKM map setting, calibrate them with map-derived features, and then integrate them into a single residual multi-task deep model in `Try 80`. The calibration methodology was: grid search over the two-ray parameters (ρ, ϕ) for the LoS branch, followed by Gaussian post-smoothing; and ridge regression with $\lambda = 10^{-3}$ over 15 (path loss) and 23 (spread) morphology/geometry features per regime for the NLoS and spread branches. Therefore, the novelty is not "inventing a new textbook propagation law", but building a practical hybrid pipeline that is both interpretable and trainable on the dataset at hand.

9 Sources

9.1 Project sources

- **Try 80 integration code:** `TFGEightiethTry80/src/priors_try80.py`.
- **Try 80 design note:** `TFGEightiethTry80/DESIGN_TRY80.md`.
- **Try 78 final result file:** `TFGSeventyEighthTry78/hybrid_out_final_dml/hybrid_eval_summary.json`.
- **Try 79 final test result file:** `TFGSeventyNinthTry79/test_eval_dml_hz_v2/eval_summary_test.json`.
- **Try 79 calibration file:** `TFGSeventyNinthTry79/test_eval_dml_hz_v2/calibration.json`.
- **Try 78 interpretation note:** `TFGSeventyEighthTry78/TRY78_LOS_DOCUMENTATION.md`.
- **Try 79 interpretation note:** `TFGSeventyNinthTry79/README.md`.

9.2 Literature sources

References

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[2] P. Kyösti et al., *WINNER II Channel Models*, IST-4-027756 WINNER II Deliverable. https://www.ist-winner.org/WINNER2-Deliverables/D1.1.2v1.2.pdf

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[4] W. Khawaja, I. Guvenc, D. Matolak, U.-C. Fiebig, and N. Schneckenberger, "A Survey of Air-to-Ground Propagation Channel Modeling for Unmanned Aerial Vehicles," *arXiv preprint arXiv:1801.01656*, 2018. https://arxiv.org/abs/1801.01656

[5] X. Cai, J. Rodríguez-Piñeiro, X. Yin, B. Ai, G. F. Pedersen, and A. Perez Yuste, "An Empirical Air-to-Ground Channel Model Based on Passive Measurements in LTE," *arXiv preprint arXiv:1901.07930*, 2019. https://arxiv.org/abs/1901.07930

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[7] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Prentice Hall. https://search.worldcat.org/es/title/Wireless-communications-%3A-principles-and-practice/oclc/318358858